MODULE 22 · SECTION 3 - STORING THE DATA

Cloud Storage: Classes, Lifecycle and Retention

Storage classes and minimum durations, Autoclass, lifecycle rules, versioning and soft delete, retention and Bucket Lock, locations and turbo replication.

SECTION 3 of 5 EXAM WEIGHT 20% MODULE 22 of 45

admin@sthithapragna.com

The class sets the price of reading.
Retention decides whether deleting is allowed at all.

Four classes, one API

CLASS	MINIMUM DURATION	RETRIEVAL FEE	READ ABOUT
Standard	None	None	Often
Nearline	30 days	Yes	Once a month
Coldline	90 days	Yes	Once a quarter
Archive	365 days	Yes	Under once a year

THE TRAP

Every class reads in milliseconds, with the same eleven nines of durability. Archive is not tape.

What a minimum duration really means

- Delete a Coldline object at day 10, and you pay for **all 90 days**.
- Early deletion charges apply to soft-deleted objects too, except in Standard.
- A lifecycle class change counts the **time already spent** in the earlier class.
- The bucket's **default class** only applies to new objects.

> REMEMBER
>
> Short-lived data belongs in Standard, however cheap the colder price looks.

Autoclass or lifecycle rules?

AUTOCLASS

Follows actual reads

- Everything starts in Standard
- 30 days unread: Nearline
- Archive as terminal: Coldline, then Archive
- A read moves it back to Standard
- No retrieval or early deletion fees

LIFECYCLE RULES

Follows age and dates

- You write the conditions
- Only moves to colder classes
- Can also delete objects
- Cheapest when the pattern is known
- Cannot set classes with Autoclass on

THE TIEBREAKER

Nobody knows how the data will be read: Autoclass. The pattern is fixed by age: lifecycle rules.

Larkspur's history bucket

```
{"lifecycle": {"rule": [
  {"action": {"type": "SetStorageClass", "storageClass": "COLDLINE"},
   "condition": {"age": 90, "matchesStorageClass": ["STANDARD"]}},
  {"action": {"type": "SetStorageClass", "storageClass": "ARCHIVE"},
   "condition": {"age": 365, "matchesStorageClass": ["COLDLINE"]}},
  {"action": {"type": "Delete"}, "condition": {"age": 2557}},
  {"action": {"type": "Delete"},
   "condition": {"isLive": false, "daysSinceNoncurrentTime": 30}}
]}}

gcloud storage buckets update gs://lk-history-prod --lifecycle-file=rules.json
```

2-5 Settlement re-runs read the last quarter, then the files go cold.

6 About seven years, then delete. The retention policy still has the last word.

HONEST LIMIT

Rules can take up to 24 hours to take effect, and actions run asynchronously after that.

How rules combine

- Conditions in one rule are joined with **AND**. Separate rules act as **OR**.
- If two rules match at once, **Delete** beats a class change.
- A class change is a Class A operation, with **no retrieval fee**.
- Transitions only go **colder**. Nothing moves an object back to Standard.

> THE PATTERN
>
> Delete waits while an object has a hold, or is still inside its retention period.

Object Versioning or soft delete?

OBJECT VERSIONING

Keeps older versions

- Overwrite or delete makes a noncurrent version
- Kept until you remove it
- Lifecycle rules tidy old versions
- Every version is billed

SOFT DELETE

An undo window

- On by default for 7 days
- Set from 7 to 90 days, or 0 to turn off
- Restores deleted objects and buckets
- Billed while it is kept

THE TIEBREAKER

'Recover last week's accidental delete' is soft delete. 'Get back the version before the bad job' is versioning.

Retention policies and Bucket Lock

- A **retention policy** blocks delete and overwrite until each object is old enough.
- **Locking** it is irreversible. Lengthen it later, but never shorten or remove it.
- **Event-based holds** start the clock when the hold is released, such as a contract end.
- **Object Retention Lock** sets retention object by object, not for the whole bucket.

> THE TRAP
>
> A locked bucket cannot be deleted, and the project gets a lien. Test the policy unlocked first.

Where the bucket lives

LOCATION TYPE	WHAT YOU GET	REPLICATION
`Region`	Copies across zones in one region	Within the region
`Dual-region`	Two regions that you choose	Default RPO 12 hours
`Dual-region, turbo`	The same two regions	RPO 15 minutes
`Multi-region`	Regions Google chooses, far apart	Default RPO 12 hours

THE SIGNAL

Turbo replication exists only on dual-region buckets. A multi-region cannot turn it on.

Sharing and encrypting

- A **signed URL** lets anyone holding it read or upload, for up to **7 days**.
- Prefer **uniform bucket-level access**, so IAM alone decides who gets in.
- A **default CMEK** key encrypts new objects. Its key ring sits in the bucket's location.
- Changing the default key does not re-encrypt old objects. **Rewrite** them.

> FOR LARKSPUR
>
> German customer files go in europe-west3, with their key ring in europe-west3 too.

Newer than most question banks

- **Rapid Cache**, formerly Anywhere Cache: an SSD read cache in a zone, for hot analytics and AI reads.
- **Rapid** storage class, in zonal **Rapid Buckets**, for the lowest latency.
- **Hierarchical namespace**: real folders, atomic folder rename, up to 8x initial QPS.
- Namespace is chosen at creation, and rules out versioning, holds and Bucket Lock.

> CURRENCY CHECK
>
> Exam questions may still say Anywhere Cache. It is the same product under its new name.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Access pattern is **unknown**	Autoclass
Data cools on a **fixed schedule**	Lifecycle rules
Nobody may delete it, **even admins**	Retention policy, then Bucket Lock
Undo an **accidental delete**	Soft delete
Survive a region, **15 minutes** of loss	Dual-region with turbo replication
Share one file, **no Google account**	Signed URL

THE PATTERN

Cheap to keep, safe to keep, quick to recover: three different questions.

> Larkspur's data scientists keep training extracts in one bucket. Some are read daily for weeks, others never again.
>
> Nobody can predict which. They want the lowest storage cost, with no rules to maintain and no retrieval charges.

What should the team configure?

A Set Nearline as the bucket's default storage class

B A lifecycle rule moving every object to Coldline at 30 days

C Turn on Autoclass for the bucket

D Set Archive as the default, and copy files to Standard on request

◆ PAUSE AND DECIDE

What should the team configure?

A Nearline default

Retrieval fees on every daily read, and a 30-day minimum.

B Coldline at 30 days

A fixed rule for an unknown pattern, and retrieval fees on reads.

C Autoclass

Follows real reads, and charges no retrieval or early deletion fees.

D Archive plus copies

A 365-day minimum, retrieval fees, and copies to manage by hand.

THE SIGNAL

Unpredictable reads, no rules, no retrieval charges: Autoclass.

> Larkspur's settlement files must be kept unaltered for seven years.
>
> The auditor wants proof that nobody, including project owners, can delete them or shorten that period.

What should the team do?

A. Turn on Object Versioning, with a rule deleting versions after seven years

B. An IAM deny policy on deletes, for everyone except a break-glass group

C. Set soft delete to its longest window, so deletions can be undone

D. Add a seven-year retention policy to the bucket, then lock it

◆ PAUSE AND DECIDE

What should the team do?

A **Versioning**
Old versions can still be deleted by anyone with permission.

B **IAM deny policy**
An administrator can change the policy, so it proves nothing.

C **Long soft delete**
Ninety days at most, and the setting can be changed.

D **Locked retention policy**
Irreversible: no delete, no overwrite, no shortening.

THE SIGNAL

Nobody, including owners, can delete or shorten it: a locked retention policy.

> Larkspur's settlement output bucket must survive the loss of a whole region.
>
> No more than 15 minutes of newly written files may be at risk. The data may be held in either of two chosen regions.

Which location should the bucket use?

A. A multi-region bucket, with the default replication settings

B. A dual-region bucket, with turbo replication turned on

C. A regional bucket, plus a nightly copy to a second region

D. A regional bucket, with soft delete kept for the full ninety days

◆ PAUSE AND DECIDE

Which location should the bucket use?

A Multi-region

Default replication allows up to twelve hours of loss, and no turbo.

B Dual-region with turbo

A fifteen-minute recovery point, backed by an SLA.

C Nightly copy

Up to a day of new files at risk.

D Soft delete

Protects against deletes, not against losing a region.

THE SIGNAL

Lose a region, fifteen minutes at most: dual-region with turbo replication.

> The nightly settlement run writes staging extracts that a job deletes after three days.
>
> To save money, an engineer put them in a Coldline bucket. The storage bill went up, not down.

What should the team do?

A. Keep the extracts in Standard, in a bucket with soft delete off

B. Move the extracts to Archive, which has the lowest price at rest

C. Turn on Object Versioning, so that each deletion is recorded

D. Add a lifecycle rule moving the extracts to Nearline after a day

◆ PAUSE AND DECIDE

What should the team do?

A Standard, no soft delete

No minimum duration, and deleted extracts stop costing at once.

B Archive

A 365-day minimum, so each three-day file costs a year.

C Versioning

Keeps even more copies, and changes nothing about the minimum.

D Nearline after a day

Lifecycle cannot move objects to Nearline from Coldline.

THE SIGNAL

Short-lived files: Standard, and no soft delete on temporary data.

Module 22 in five lines

- Nearline **30**, Coldline **90**, Archive **365** days. All read in milliseconds.
- **Autoclass** for unknown reads. **Lifecycle rules** for a known schedule.
- **Soft delete** undoes deletes. **Versioning** keeps old versions.
- A **locked retention policy** stops anyone deleting or shortening.
- Turbo replication: **dual-region only**, a 15-minute RPO.

COMING UP

Next: module 23, BigQuery tables: partitioning, clustering and storage.